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Studierichting: Informatica

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$$9^x + 3^{x-3} = 1 + 27^{x-1}$$

$$3^{2x} + 3^{x-3} - 3^{3x-3} - 1 = 0$$

$$(3^x)^2 + 3^x \cdot 3^{-3} - (3^x)^3 \cdot 3^{-3} - 1 = 0$$

$$\text{Laat } X = 3^x$$

$$X^2 + 3^{-3}X - 3^{-3}X^3 - 1 = 0$$

$$-X^3 + 27X^2 + X - 27 = 0$$

$$-X(X^2 - 1) + 27(X^2 - 1) = 0$$

$$(X^2 - 1)(27 - X) = 0$$

$$\Rightarrow X = 1, -1, 27$$

$$\Rightarrow 3^x = 1, -1, 27$$

$$\Rightarrow \boxed{x \in \{0, 3\}}$$

References